

TMS Milkrun — User Manual

Transport Management System (Milkrun) · ระบบบริหารจัดการการขนส่ง

Version 1.0 · Bilingual (English / ไทย)

This manual is written in Markdown so it can be converted to PDF. To produce a PDF, open it in any Markdown viewer and “Print → Save as PDF”, or run a converter such as `pandoc "User Manual.md" -o "User-Manual.pdf"`.

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1. Overview

TMS Milkrun is a web application for planning and running **milkrun** transport operations — repeated collection/delivery loops between a depot and a set of supplier, plant, warehouse, and customer locations.

Key capabilities

- **Auto Route** — a VRPTW optimizer (nearest-neighbour + 2-opt + iterated local search) honouring **m³ and kg** capacity and **time windows**, optimizing for **lowest cost / shortest distance / balanced load**, planning on **real Mapbox road distances and drive times**.
- **Multi-depot milkrun** — each supplier delivers to a **destination plant**; the optimizer builds a loop that starts and ends at that plant, so a truck only carries goods bound for one plant. Each supplier has a **rounds/day cadence**.
- **Day / night shift** — plan either shift; the shift sets the departure time, the pickup windows used, and the labor/fuel rates.
- **Rate-card costing** — each transporter has a detailed **rate card** (labor + OT, fuel, km-allowance, drop-points, daily fixed, admin %); the plan is priced as a real daily cost, with a **transporter comparison** (who is cheapest).
- **Interactive editing** — drag to reorder stops, reassign a stop, lock a route, and run “what-if” fleet scenarios with before/after savings.
- **Operations** — trip status workflow, proof of delivery, delay tracking, printable driver route sheets.
- **Finance** — Thai VAT 7% / withholding tax 1%, invoices, statements, manifest, and an **Excel simulation report**.
- **Bilingual (TH/EN)**, **light & dark themes**, **role-based access** (admin/dispatcher/viewer), an **activity log**, and **CSV import**.

- **Real database** — Neon Postgres via a small API server.

The sample dataset is a real **AISIN inbound-milkrun network**: 7 plants, 15 supplier lanes (m³/kg, pickup windows, 1–3 rounds/day), a Yusen 6W/10W fleet, and five transporter rate cards. Running Auto Route reproduces an 8-route milkrun at roughly ₪33k/day.

1a. Milkrun workflow (end to end)

1. **Master data** — add **plants** (Delivery Locations with type *Plant*) and **suppliers** (type *Supplier*); on each supplier set its **Delivery plant**, **Pickups per day**, demand (m³/kg), and pickup window. Add **Trucks** (6W/10W with m³ + kg capacity) tied to a **Transport Partner**, and give each partner a **Milkrun rate card** per truck type.
2. **Choose the run** — in **Route Planner** pick **Optimize for** (Lowest cost is the default), the **Shift** (Day/Night), and optionally a **weekday** and a **what-if** fleet (exclude trucks).
3. **Auto Route** — the optimizer groups suppliers by plant, fetches the real Mapbox road matrix, builds and improves each plant loop, and prices every route with the partner's rate card. The plan-total card shows daily cost, distance, and CO₂.
4. **Review & adjust** — inspect each route card (utilization, xN rounds/day, cost, in-window/late badges); drag to reorder or reassign stops, or lock a route and re-run.
5. **Cost & compare** — open **Cost Summary** for the fixed/variable breakdown and the **transporter comparison**; export the **Excel simulation report**.
6. **Operate** — dispatch routes, record proof of delivery, then bill from the plan in **Billing & Payments**.

2. Installation & Setup

Requirements

- Node.js 20+ (developed on Node 22)
- A Neon Postgres connection string (or any Postgres database)
- A Mapbox access token (free at <https://account.mapbox.com/access-tokens/>)

Configure environment

Copy `.env.example` to `.env` and fill in:

```
VITE_MAPBOX_TOKEN=pk.your_mapbox_token
DATABASE_URL=postgresql://user:password@host.neon.tech/neondb?sslmode=require
```

- `VITE_MAPBOX_TOKEN` shows the map. It can also be entered later in **Settings**.
- `DATABASE_URL` is used **only by the API server** and is never exposed to the browser.

Install & run

```
npm install
npm run dev      # starts BOTH the API server (:3001) and the web app (:5173)
```

Open <http://localhost:5173>.

- `npm run web` — run only the web app.
- `npm run server` — run only the API server.
- `npm run seed` — insert the sample dataset into Neon (`--force` to wipe + reseed).
- `npm run build` — type-check and produce a production build.

The sample dataset lives in `server/seed.mjs` and is inserted into Neon by the API server on first run (when the database is empty). There is **no mock data in the browser** — the app always loads real records from the database.

3. Navigating the App

The **sidebar** on the left groups pages:

- **Top:** Dashboard, Route Planner, Cost Summary, Billing & Payments
- **Operations:** Operations, Visual Truck 3D, Incident Log
- **Master Data:** Delivery Locations, Products, Trucks, Drivers, Transport Partners
- **Settings**

Language switch — buttons at the bottom of the sidebar (**EN / ไทย**). The choice is saved to the database.

Alert bell (top of the sidebar) — a live count of things needing attention; click an alert to jump straight to the relevant page. See §12.

4. Master Data

All master pages support **Add**, **Edit** (pencil), and **Delete** (trash), and most support **Export CSV**.

4.1 Delivery Locations

Each location has:

Field	Meaning
Code	Short identifier (e.g. <code>SUP-01</code>)
Name (EN) / Name (TH)	Bilingual names
Type	Supplier / Plant / Warehouse / Customer
Zone	Region grouping (e.g. Chonburi, Rayong) — filterable
Latitude / Longitude	Validated coordinates
Demand (m³/day), (kg/day)	Volume and weight to move
Service Time (min)	Handling time / loading time at the stop
Delivery Window	Earliest / latest time (HH:MM)
Delivery plant (suppliers)	The Plant location this supplier delivers to — turns on milkrun mode : Auto Route loops the truck back to this plant. Leave as <i>Global depot</i> for single-depot planning.
Pickups per day (suppliers)	How often this supplier is collected (1–3). The milkrun loop runs at the cadence of its most-frequent supplier.
Delivery Days	Which weekdays the stop is served (empty = every day) — drives multi-day planning

Milkrun tip: create your plants first (type *Plant*), then set each supplier's **Delivery plant**. As soon as one supplier has a

plant, Auto Route switches to multi-depot milkrun and groups suppliers by plant.

Coordinate validation: latitude must be -90...90 and longitude -180...180. Coordinates outside Thailand raise a soft warning so you can double-check. Tip: copy “lat, lng” straight from Google Maps.

Use the **Zone filter** and search box above the table to narrow the list.

4.2 Trucks

Plate number, type (4W / 4WJ / 6W / 10W / Trailer, with capacity presets), transport partner, **capacity (m³ + kg)**, **rounds per day**, fixed cost per round, and cost per km. The assigned **driver** is shown automatically.

Assignment mode (milkrun): each truck is either **Dynamic** (the optimizer assigns its stops each plan) or **Fixed** — a **cyclic route** where you pick the stops the truck always runs. Fixed trucks keep their loop every plan (cyclic rotation, short lead-time, stable communication); dynamic trucks absorb the rest.

4.3 Drivers

Code, EN/TH name, license number and class (e.g. ๓.2 / ๒.2), phone, and the **assigned truck**. Driver names then appear on route cards, route sheets, and the manifest.

4.4 Transport Partners

Contact details plus a **Rate Card & Terms** section:

- **Rate / km** — negotiated THB per km (leave 0 to bill by each truck’s own cost)
- **Rate / trip** — flat THB per round
- **Min. Charge** — invoice floor
- **Credit (days)** — payment terms, used for invoice due dates

Milkrun rate card (per truck type) — the detailed cost used to price milkrun routes. For each truck type (e.g. 6W, 10W) enter: labor ฿/hr, OT ฿/hr, ฿/drop, ฿/km allowance, trip fee, fuel km/L, diesel ฿/L, other ฿/day, and admin %. Night rates can differ (better fuel economy off-peak). A route’s daily cost is **(rounds × trip + other) × (1 + admin)**, where trip = labor (+OT past 8h) + fuel

- allowance + drop-points + trip fee. Leave a type blank to fall back to the simple Rate/km above. Because every partner can have its own card, the Cost Summary can show **which transporter is cheapest for the same plan**.

and **Bank Details** (bank, account number, account name) used by the **Bank Payment File** export (§9).

4.5 Products

Optional catalogue of goods with physical **dimensions (W × L × H)** and **weight**, linked to a supplier location. Used by the Visual Truck 3D load planner (§11).

5. Route Planner

The heart of the system. It shows a **map** on the left and a **route list** on the right.

5.1 Auto Route

Set two controls first (both are saved):

- **Optimize for** — **Lowest cost (฿)** (the default: cheapest total delivery cost, so the cheaper cost/km truck takes the longer route and fewer trucks are used when that saves money), **Shortest distance** (fewest total km), or **Balanced load** (keeps

cost low while spreading load evenly).

- **Shift — Day** (departs 08:00, uses each stop's day pickup window and day rates) or **Night** (departs 20:00, uses the night window — windows crossing midnight like 20:00–05:00 are handled — and the transporter's night rates).

In **milkrun mode** (any supplier has a Delivery plant) Auto Route groups suppliers by plant and builds a loop per plant. Then click **Auto Route**. The optimizer:

1. builds one trip per truck round with a **time-window + capacity nearest-neighbour** pass, respecting **both m³ and kg** and filling **rounds per day**,
2. sequences each trip with time-window-feasible **2-opt**,
3. runs a **metaheuristic optimization pass** (iterated local search with ruin-and-recreate) — it **relocates** and **swaps** stops between routes and **exchanges trucks** (so the cheaper cost/km truck takes the longer route, and fewer trucks are used when that is cheaper) to **lower total cost and even out load**, repeatedly disrupting and rebuilding part of the plan to escape local optima, accepting a change only if every route stays within capacity **and** its time windows, and **reinserts** any unassigned stop that now fits. The result is **repeatable** (the same input always gives the same plan), then
4. (if a Mapbox token is set and **Snap routes to real roads** is on) plans on a **real Mapbox road matrix** — cost and stop order use true road **distances** and the arrival times / **time-window checks use true road travel times** (real durations, not a constant speed) — then **snaps each route polyline to the roads** and re-prices it with real distance.

Each **route card** shows truck, driver, partner, round, trip status, volume/ weight utilization, distance, duration, cost, and estimated CO₂.

Time windows are enforced. Arrival times run from the depot departure clock (**Settings → Depot Departure**, default 08:00). A stop is only assigned to a truck if it both **fits capacity** and can be **reached within its delivery window** — the truck waits if it arrives early, and a stop that can't be reached in time is left unassigned. Fixed cyclic routes keep all their stops but flag any late arrival. Selected-route stops show an **In window / Late Nm** badge.

Stops that don't fit any truck **or can't meet their time window** appear under **Unassigned Stops** — add trucks, rounds, or widen the window.

5.2 The map

- Style switcher: **Streets / Satellite / Dark / Traffic**, plus **3D** tilt with building extrusion.
- Location dots are coloured by type; the depot is marked separately.
- **Click a route card** to highlight it on the map: the map flies to it, other routes dim, an animated “flow” line and numbered stop badges appear, and a truck marker animates along the road.

5.3 Editing a plan (manual overrides)

Click a route card to expand its **stop list**, then:

- **Reorder stops** — drag a stop up or down. The route re-computes distance, duration, ETA, and cost, and re-snaps to roads.
- **Reassign a stop** — use the **Move to...** dropdown on a stop to send it to another truck. Both routes re-compute. (Over-capacity is allowed but the utilization % turns red to flag it.)
- **Lock a route** — click **Lock**. A locked route keeps its stops and its truck/round slot the next time you run **Auto Route** — only the rest of the network is re-optimized. Click **Unlock** to release it.

5.4 What-if & savings

- **Fleet — what-if** (expandable panel): untick any truck to **exclude** it from the next Auto Route — useful for “what if this truck is in for service?”.
- After any re-plan, a **Savings vs previous plan** banner shows the change in cost, distance, and CO₂ (green = reduced).

5.5 Multi-day planning

Locations can carry a **delivery-day schedule** (which weekdays they need service — set in Delivery Locations; empty = every day). In the Planner, the **Plan day** panel lets you:

- pick a weekday (or **Every day**) — **Auto Route** then plans only the stops scheduled for that day, and
- see a **Weekly demand** overview (stops and m³ per weekday) to balance the week at a glance.

5.6 Export — simulation report

Export Excel (on the Planner and Cost Summary) produces a styled multi-sheet **simulation report**: a Summary of KPIs, the **Routes** (stop-by-stop), a **Cost** sheet (per route with destination plant, rounds/day, fixed vs variable, total), the **Transporter comparison**, and the master data. Use it to share the plan or to compare scenarios offline.

6. Operations (Trip Execution)

For running the plan on the day.

- **Trip status workflow**: Planned → **Dispatch** → **Start** (in transit) → **Complete** (with **Reopen** to revert). Status is shown as a badge.
- **Departure time**: set each route's planned departure clock; planned arrival times per stop are computed from it.
- **Proof of Delivery (POD)**: click **Record delivery** on a stop to capture status (delivered / failed / pending), **actual arrival time**, receiver name, a note, and a **photo**. A progress bar tracks delivered stops.
- **ETA / delay**: once an arrival time is recorded, each stop shows a delay badge — **On time**, **Late N min**, or **Early N min** — versus the planned ETA.
- **Route Sheet**: click to open a **printable driver sheet** for one route (stops, ETAs, windows, demand, signature lines).
- **Dispatch Manifest**: the header button prints a one-page manifest of the whole plan (§10).

7. Incident Log

Record operational exceptions: date, **type** (breakdown / delay / accident / damage / other), **severity** (low / medium / high), affected truck, description, and a **Resolved** flag. High-severity open incidents raise an alert. Export to CSV.

8. Cost Summary

A pivot of the current plan's cost, grouped **By Partner / By Truck / By Route**. Costs use each transporter's **rate card** when set, so the **Fixed** column is labor + trip fee + daily other and **Variable** is fuel + km-allowance + drop-points (both include admin); otherwise it falls back to the truck's simple fixed + per-km cost. It also shows THB per m³, the **daily total**, and a **x22-day monthly estimate**.

Transporter comparison — below the pivot, the same plan is re-priced under **every** transporter's rate card and ranked cheapest-first, with each one's daily and monthly cost and its gap versus the cheapest (e.g. *Yusen +1.7%*, *Karitsu +37%*). This answers “who should we award this milkrun to?” at a glance. Everything exports to the Excel simulation report.

8a. Milkrun Analytics

The **Analytics** page turns the current plan into milkrun KPIs and charts (CSV export included). Top tiles cover the four principles — **Cyclic Rotation** (share of fixed cyclic routes), **Short Lead-Time** (avg/max route duration), **Loading Efficiency** (volume utilization), and **Flexible to Change**. Below:

- **Truck Routing** — routes, stops, distance, and utilization per truck, with fixed/dynamic mode.
- **Time Windows** — % of stops whose planned arrival falls within their window, plus on-time / late / early from PODs.
- **Load Optimization** — m³ and kg utilization per route.

- **Returnable Packaging** — returnable (wooden/plastic pallets that must cycle back to the depot) vs. one-way, derived from the Products master.
- **Flexibility & Communication** — delivery completion, failed deliveries, open incidents, and CO₂.

9. Billing & Payments

Turn a plan into invoices.

- **Create Billing from Plan** — generates one billing record **per transport partner**, applying that partner's **rate card** (THB/km and/or THB/trip), **minimum charge**, and **credit-day** due date.
- Each invoice carries **VAT 7%** and **withholding tax 1%** (the Thai rate for transport services). **Net payable = (subtotal + fuel surcharge) + VAT – WHT.**
- **Workflow:** Draft → **Approve** → **Mark Paid** (records the paid date).
- **Edit** an invoice to adjust invoice number, fuel surcharge %, or notes — a live preview recalculates the totals.
- **KPI tiles:** Total Billed, Outstanding, **Overdue** (red), Paid This Month.
- **Export Excel** — a billing register with the full tax breakdown.
- **Bank Payment File** — export a bank bulk-transfer CSV of all outstanding invoices, grouped by partner, using each partner's **bank details** (set in Transport Partners). Columns: recipient name, bank, account no., amount, reference (invoice numbers), and payment date.

Document buttons on this page are covered next.

10. Printed Documents

All documents open in a print window with a “**Print / Save as PDF**” button. This route renders Thai text perfectly and, when saved, produces a real PDF. Set your company header in **Settings → Company** first.

Document	Where	Contents
Tax Invoice (ใบกำกับภาษี)	Printer icon on each Billing row	Company letterhead + Tax ID, Bill-To, line items, VAT/WHT, net payable, and amount in words (Thai or English)
Monthly Statement	“Monthly Statement” on Billing header → pick partner + month	All invoices for that partner/month, totals, and outstanding balance
Dispatch Manifest	“Dispatch Manifest” on Operations header	All routes with truck, driver, partner, stops, m ³ /kg/km, status, and totals
Route Sheet	“Route Sheet” per route on Operations	Per-truck driver sheet with stop sequence, ETAs, and signature lines

Dates on documents follow the selected language — Buddhist era (e.g. 2569) in Thai, Gregorian in English.

11. Visual Truck 3D & Products

The **Products** master (see §4.5) defines cargo items with physical dimensions and weight per supplier.

Visual Truck 3D provides an interactive 3D view of how cargo loads into a truck body, helping visualize space utilization for a route. Select a truck/route to see the load arrangement. (This complements the numeric m³/kg utilization shown on route cards.)

12. Dashboard & Alerts

Dashboard — KPI tiles (active locations, trucks, planned routes, total distance, daily cost, **total CO₂**, average utilization) plus bar charts for per-route utilization and cost by partner.

Alert Center (bell in the sidebar) — a live badge and dropdown listing:

- overdue invoices,
- unassigned stops in the plan,
- open / high-severity incidents,
- failed deliveries,
- trips in progress,
- drivers without an assigned truck.

Click any alert to navigate to the relevant page.

13. Settings

- **Language** — English / ไทย (also on the sidebar).
- **Mapbox Access Token** — required to display the map.
- **Depot (Route Origin)** — name and coordinates; the planning start/end point.
- **Average Speed** and **Snap routes to real roads** toggle.
- **Appearance** — **Light / Dark** theme (also toggled by the sun/moon icon at the bottom of the sidebar). The choice is saved to the database.
- **Access Role** — **Admin / Dispatcher / Viewer** (see §13.1).
- **Fuel & Emissions** — diesel price (THB/L), fuel economy (km/L), and CO₂ per litre. **Apply fuel price to all trucks' cost/km** recomputes each truck's cost from the current fuel price.
- **Company (for documents)** — company name, Tax ID, and address printed on invoices, statements, and manifests.
- **Activity Log** — a change history of who did what, when (see below).
- **Data Management** (admin only) — **Reset to Sample Data** or **Clear All Data**.

13.1 Sign in, roles & access control

The app requires a **login**. Each user has a role, and the role is enforced by the server (not just hidden in the UI):

Role	Can do
Admin	Everything, including settings, data management, and reset
Dispatcher	Plan routes, run operations, and create billing — but not edit master data or system settings
Viewer	Read-only — no create/edit/delete anywhere (server rejects writes); exports still allowed

Default accounts are created on first run — **change these**: `admin/admin` , `dispatcher/dispatcher` , `viewer/viewer` (override with `ADMIN_PASSWORD` etc.). Sessions use an httpOnly cookie; **Settings** → **Log out** ends the session. When a role lacks a capability, its buttons (Add, Import, Auto Route, edit/delete) are hidden, and the API rejects the action if attempted directly.

13.2 User management (admin only)

Admins get a **Users** page (Administration section of the sidebar) to **add** users (username, role, password), **change** a user's role or reset their password, and **delete** users. Safeguards prevent deleting your own account or removing the last admin. Passwords are stored script-hashed; the plaintext is never shown or returned.

13.3 Activity Log

Every create / update / delete of master data, every billing action, and every settings change is recorded with a timestamp and the acting role. The **Activity Log** in Settings shows the most recent entries; the full log is persisted in the database.

13.4 CSV import

Delivery Locations supports **Import CSV** (admin/dispatcher). Use the same column headers as the CSV export (`Code`, `Name`, `NameTH`, `Kind`, `Zone`, `Lat`, `Lng`, `DemandM3`, `DemandKg`, `ServiceMin`, `WindowStart`, `WindowEnd`, `Active`). Rows whose `Code` matches an existing location **update** it; others are **created**. Coordinates are validated and invalid rows are skipped.

14. Data & Persistence (Neon Postgres)

State is stored in **Neon Postgres**, not the browser.

- A small Express API (`server/index.mjs`) owns the `DATABASE_URL` and serves `GET /api/state` (load everything), `PUT /api/state` (save everything), and `POST /api/seed` (re-seed).
- The canonical dataset lives in `server/seed.mjs` ; the server inserts it into Neon on first run. The browser store starts **empty** and loads real records from the database — no mock data ships in the client.
- The frontend hydrates from the API on startup and **auto-saves** (debounced) on every change. Saves are serialized so concurrent writes never collide.
- **Reset to Sample Data** (Settings) calls `POST /api/seed` , which re-seeds the database, then reloads.
- Tables: `partners` , `trucks` , `drivers` , `locations` , `products` , `billings` , `pods` , `incidents` (each `id` , `doc jsonb`) plus a `singletons` table for `settings` and `plan` .

Because everything is in the database, opening the app on another machine (with the same `DATABASE_URL`) shows the same data.

15. Import / Export

- **Excel export** — Route Planner, Cost Summary, and Billing pages produce styled `.xlsx` workbooks (localised to the current language).
 - **CSV export** — Delivery Locations, Drivers, and Incident Log.
 - All exports respect the current language and number formatting.
-

16. Troubleshooting & FAQ

The map is blank. Add a Mapbox token in **Settings**, then hard-refresh (Cmd/Ctrl+Shift+R). Without a token, planning still works using straight-line road estimates.

Routes are straight lines / distances look off. Ensure a Mapbox token is set and **Snap routes to real roads** is on, then run **Auto Route** again. Existing plans self-upgrade to real roads when the Planner opens with a working token.

Data didn't save. Confirm the API server is running (`npm run server`) and that `DATABASE_URL` is correct — check `http://localhost:3001/api/health` returns `{"ok":true}` .

A stop won't fit any truck. It appears under **Unassigned Stops** — add a truck, add rounds, or increase a truck's capacity.

Thai text is garbled in a document. Use the **Print / Save as PDF** button in the document window (it uses the browser's Thai font); avoid third-party PDF tools that don't embed Thai fonts.

17. Glossary

Term	Meaning
Milkrun	A fixed loop collecting/delivering between a depot and multiple stops
CVRP	Capacity-Constrained Vehicle Routing Problem — the optimizer's model
Round	One trip of a truck in a day; a truck may run several rounds
m³ / kg utilization	How full a truck is by volume / weight (100% = full)
POD	Proof of Delivery — arrival time, receiver, note, photo
VAT (ภาษีมูลค่าเพิ่ม)	Value Added Tax, 7% in Thailand
WHT (หัก ณ ที่จ่าย)	Withholding tax, 1% for transport services
Net Payable (ยอดชำระสุทธิ)	(subtotal + fuel surcharge) + VAT – WHT
Depot (ศูนย์กระจายสินค้า)	The origin/return point for all routes
Locked route	A route excluded from re-optimization

© TMS Milkrun. Sample company data (AISIN Thailand) is illustrative.